Target Recognition for SAR Images Enhanced by Polarimetric Information

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***Abstract*— Target recognition for synthetic aperture radar (SAR) images has been a longstanding hotspot. However, using polarimetric information to enhance recognition performance is under-researched. In this article, we develop a visualization approach to analyze and highlight the contributions of polarimetric elements and propose a simple polarimetric correlation feature for target recognition. In the visualization method, a channel-wise convolutional structure is well developed, which serves as a proxy of the polarimetric elements. By using the gradients of the target class flowing into each developed convolutional channel with normalization, we obtain activation maps indicating the contribution of each polarimetric element. Then, a comprehensive quantitative evaluation of polarimetric element contribution is also performed. We demonstrate again that polarimetric information maintains significant advantages over single-polarization intensity, and the correlations between cross- and co-polarization emerge as key components for target recognition. Exactly inspired by these insights, the straightforward feature is defined as the correlation between the target and typical scatterers, seamlessly fusing intensities and correlations. This polarimetric correlation feature aptly encapsulates the physical scattering of the target, shows a clear mapping relationship with the geometric structure, and captures the differences between categories, achieving intracategory compactness and intercategory separation. Experimental results on both synthetic and real datasets underscore the effectiveness and superiority of this proposed feature.**

***Index Terms*— Polarimetric element contribution, polarimetric information, synthetic aperture radar (SAR), target recognition.**

I. Introduction

THE swift progression of synthetic aperture radar (SAR) systems, coupled with the escalating demand for large-scale data processing, has thrust the intelligent interpretation of SAR images into the limelight as a vanguard issue in SAR applications, garnering widespread attention. As a crucial component to realizing intelligent interpretation of SAR images, target recognition has significant prospective applications in various fields, such as ocean management, environmental conservation, and battlefield surveillance [1].

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Over the past few decades, numerous algorithms have been proposed to tackle the challenge of SAR target recognition. The literature on SAR target recognition generally distinguishes between traditional methods and more recent strategies reliant on deep learning.

Early SAR target recognition methodologies were almost template-based methods [2], [3], which required extensive storage of templates and used a designed matching technique. The efficacy was largely constrained by the number of templates and the challenge of obtaining them. Consequently, researchers introduced model-based methods [4], [5], which simulated the scattering process with assistance from computer-aided design models and electromagnetic simulation software. Nevertheless, these simulations demanded significant computational resources, and their results often deviated from the actual SAR targets. The advent of machine-learning approaches led to the application of various algorithms, including support vector machines (SVMs) [6], k-nearest neighbor (KNNs) [7], and Adaboost [8], to the issue of SAR target recognition. Besides, Mishra validated and compared the recognition performance of PCA and LDA [9]. Thiagarajan et al. [10] first adapted sparse representation to SAR target recognition. Subsequently, dictionary learning-based methods [11], [12] were proposed to learn more compact and representative embeddings. In addition to the aforementioned classifiers, multiple feature representations have been meticulously designed for SAR image target recognition. For instance, Srinivas et al. [13] used the wavelet-transformed features of images, and Dong et al. [7] employed the monogenic transformation of images as features. Moreover, attribute scattering center-based methods [14], [15], [16], [17] exploited the scattering center to simulate the target's physical optics and geometric diffraction, estimate the target's position and geometric structure size, and accomplish target recognition. However, in practical scenarios, scattering centers are susceptible to coupling, and independent scattering centers are often challenging to acquire. Building on optical image characteristics, researchers have proposed a series of statistical features compatible with SAR images, which predominantly include the multilevel local histogram [18], ratio detector-based feature [19], and SAR-SIFT [20]. Clemente et al. exploited pseudo-Zernike moments [21] and Krawtchouk moments [22] to extract features with diversities and invariant properties. Although these global or local statistical features are rotation-invariant, they are easily influenced by variables such as changes in incident angle and clutter, and their information is confined to the image domain.

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In recent years, deep-learning techniques have revolutionized numerous fields, gradually cementing their status as valuable tools for SAR target recognition [23], [24]. Among these techniques, the convolutional neural network (CNN) stands out as a prime example [25]. Chen et al. [26] proposed all-convolutional networks (A-ConvNets) that exclusively contain sparsely connected layers, thereby dispensing with fully connected layers. Shang et al. [27] introduced a network structure known as M-net, devised for vehicle target recognition. Furthermore, Yu et al. [28] developed a complex-valued fully CNN (CV-FCNN), upon which Tan et al. [29] built to introduce 3-D convolution (CV-3D-CNN). Deep neural network-based methods can attain superior classification accuracy; however, they are prone to overfitting due to the large number of network parameters and limited quantity of training samples.

The approaches previously described largely rely on the intensity information of SAR images and were evaluated under rudimentary conditions. Current methods using the moving and stationary target acquisition and recognition (MSTAR) dataset achieve classification accuracies above 95%, sometimes exceeding 99%. However, refining classification accuracy on this rudimentary dataset may yield negligible returns. Furthermore, these ostensibly superior methods may significantly underperform or even fail when applied to complex environments, such as those with low signal-clutter-ratios.

Polarization, the spatial variation descriptor of the electric field vector in electromagnetic waves, could substantially enhance SAR target recognition when fully exploited. Since the establishment of ESA ENVISAR/ASAR in 2002, a series of polarimetric SAR satellites have been launched, providing sufficient system and data support for the development of polarization theory. Polarization decomposition, as evidenced in numerous [30], [31], [32], [33], [34], [35], is a key instrument in extracting polarimetric features by dividing the polarimetric coherence matrix into multiple independent components according to physical scattering models. Han and Shao [36] leveraged Yamaguchi decomposition [35] for full polarimetric SAR image classification. In contrast, Paladini et al. [37] focused on classifying man-made targets through invariant coherency-matrix eigenvector decomposition of polarimetric SAR images. However, polarization decomposition is frequently employed for distributed target classification (land-over classification). Based on scattering characteristics and polarization mechanism, many polarimetric features, such as similarity coefficient [38], [39] polarization cross-entropy [40], neighborhood polarimetric covariance matrix [41], and polarimetric correlation pattern [42], [43] have been proposed in terms of polarimetric SAR target detection. For polarimetric SAR target recognition, Yi et al. [44] assessed the efficacy of complex-valued nonneighborhood ResNets [45], [46] for vehicle recognition, while Li et al. [47] suggested coupling monogenic signal and complex-value nonlocal network (MS-CVNN), utilizing deep features from the complex-valued network to compensate for the limitations of artificial features.

The application and contribution of polarimetric information in existing literature are primarily concentrated on SAR target detection, with sparse studies focusing on SAR target recognition. At present, polarimetric information is underresearched, the role of polarimetric features remains unclear, and quantitative assessments are lacking. Hence, it is both necessary and urgent to investigate the contribution of polarimetric information to SAR target recognition.

This article emphasizes the role of polarimetric information in SAR target recognition. We develop a visualization approach to interpret network responses to specific inputs and highlight the contributions of polarimetric elements by imposing a channel-wise convolutional structure as the element proxy and adapting the gradient-weighted class activation mapping (Grad-CAM) [48] to a channel-level mapping tool. This approach is applied to both A-ConvNets and CV-FCNN, resulting in their extension models named contribution-visible A-ConvNets (cvA-ConvNets) and CV-FCNN (cvCV-FCNN), respectively. Then, a comprehensive quantitative evaluation of polarimetric information demonstrates again that polarimetric information outperforms single-polarization intensity significantly. Among the elements of the polarimetric covariance matrix, the correlations between cross- and co-polarization are key components. Furthermore, inspired by these insights, we propose a simple polarimetric feature, which is defined as the correlation between the target and typical scatterers. This proposed feature has a clear mapping relationship with the target geometry and shows strong discriminability across categories, exhibiting substantial potential in SAR target recognition.

Our main contributions are summarized as follows.

1) We develop a new visualization method to highlight the contributions of polarimetric elements by using a designed channel-wise convolutional structure as the element proxy and adapting Grad-CAM with channel normalization. Compared with the original Grad-CAM, our visualization method not only highlights the important regions in the image (which the original Grad-CAM does), but also points out the polarimetric channels of more interest. With this approach, we provide a quantitative evaluation of polarimetric information's contribution to SAR target recognition for the first time.
2) Inspired by the quantitative evaluation conclusions, we propose a simple polarimetric feature that adequately encapsulates the physical scattering of the target and correlates strongly with the geometric structure. In this polarimetric feature space, samples show strong discriminability across categories, achieving intraclass compactness and interclass separation. The efficacy and superiority of the proposed feature are demonstrated in experiments involving synthetic and real datasets.

The rest of this article unfolds as follows. Section II surveys the foundational works, including A-ConvNets, CV-FCNN, and Grad-CAM. Section III delineates the proposed contribution visualization approach, its application to A-ConvNets and CV-FCNN, and the polarimetric feature. Section IV offers the experiments and ensuing discussions, followed by the conclusion in Section V.

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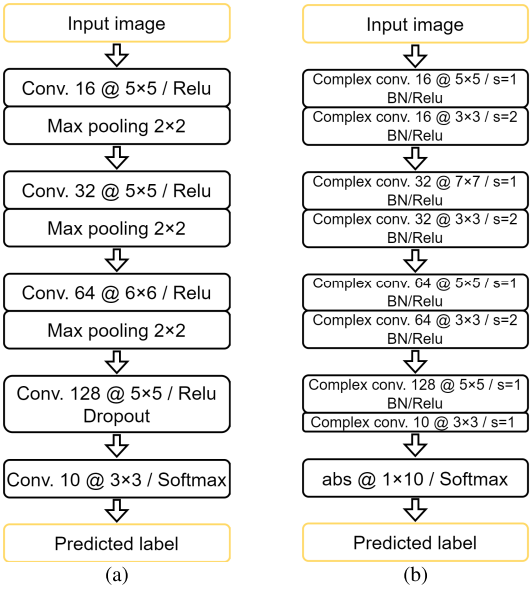


Fig. 1. Overall architecture of (a) A-ConvNets and (b) CV-FCNN. The A-ConvNets is composed of five trainable layers. The convolutional layers are represented as "Conv. (number of feature maps) @ (filter size)." The CV-FCNN consists of five blocks, containing eight trainable layers. The complex convolutional layers are represented as "Complex conv. (number of feature maps) @ (filter size) / s=(stride step size)."

II. Background

A. A-ConvNets

Aiming at the problem of SAR image data limitation, Chen et al. [26] proposed a new architecture network with fewer degrees of freedom named A-ConvNets. All layers of the A-ConvNets take the form of sparse connection rather than full connection, which allows the A-ConvNets to successfully avoid the overfitting issue associated with small training sets.

A-ConvNets consists of five blocks, and each block contains a trainable layer, as shown in Fig. 1(a). The first three blocks are composed of the convolutional layer, activation layer (ReLU), and pooling layer in sequence. The dropout layer is imposed in the fourth block, and the softmax layer is used to generate the final prediction label in the fifth block. Due to the use of local connectivity and weight sharing in convolutional layers, the number of free parameters to be learned is significantly reduced. The nonlinearity in activation layers and pooling layers provides the possibility for the powerful expressive ability of the network, and the dropout layer is used as a regularization technique to reduce overfitting.

B. CV-FCNN

To make full use of both magnitude and phase information, CV-CNN has been proposed for SAR image recognition. Yu et al. [28] developed CV-FCNN where only complex convolutional layers are used, as shown in Fig. 1(b). The purpose of replacing the fully connected layers in CV-FCNN is to prevent overfitting, which is the same as the similar operation in A-ConvNets. However, the pooling operation is also abnegated to avoid complex pooling operations further.

Compared with A-ConvNets, CV-FCNN mainly has the following differences. First, the convolution operation of CV-FCNN is defined on the complex field, that is, both the convolution kernel and the convolution input are complex numbers. However, the convolution operation of A-ConvNets is defined in the real-value field. Then, the pooling layers are replaced by the complex convolutional layers of size 3×3 with stride step size 2 in CV-FCNN, since the authors think that the pooling operation is too expensive. Finally, complex numbers are inconvenient to perform softmax operation, so magnitude operation is employed before the softmax, which is denoted as abs in the last layer in Fig. 1(b).

C. Grad-CAM

Grad-CAM uses the gradients of any target concept, flowing into the final convolutional layer to produce a coarse localization map highlighting the important regions in the image for predicting the concept. To obtain the class-discriminative location map Grad-CAM $L^c_{\text{Grad-CAM}} \in \mathbb{R}^{u\times v}$ of width u and height v for any class c, the gradient of the score of class c, y^c (before the softmax), with respect to feature maps A^k of convolutional layers, that is, $(\partial y^c/\partial A^k)$, is computed. These gradients flowing back are global-average-pooled to obtain the neuron importance weight α^c_k

$$\alpha^c_k = \overbrace{\frac{1}{Z}\sum_i\sum_j \underbrace{\frac{\partial y^c}{\partial A^k_{ij}}}_{\text{gradient via backprop}}}^{\text{global average pooling}}. \tag{1}$$

The weight α^c_k represents a partial linearization of the deep network downstream from A and captures the 'importance" of feature map k for a target class c.

The authors perform a weighted combination of forward activation maps and follow it a ReLU to obtain:

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\underbrace{\left(\sum_k \alpha^c_k A^k\right)}_{\text{linear combination}}. \tag{2}$$

Notice that this results in a coarse heat-map of the same size as the convolutional feature maps. A ReLU is applied to the linear combination of maps because the authors are only interested in the features that a *positive* influences on the class of interest, that is, pixels whose intensity should be increased to increase y_c. Negative pixels are likely to belong to other categories in the image. As expected, without this ReLU, localization maps sometimes highlight more than just the desired class and achieve lower localization performance.

Although the authors focus on explaining decisions the network possibly makes in the original article [48], Grad-CAM is very generic and can be used to visualize any activation in a deep network. Besides, unlike other approaches, Grad-CAM applies to a wide variety of CNN model families.

III. Methodology

In this section, we initially provide a succinct introduction to polarimetric SAR data, followed by an overview of the proposed visualization methodology that offers insights into the "importance" of the input polarimetric channels. Subsequently,

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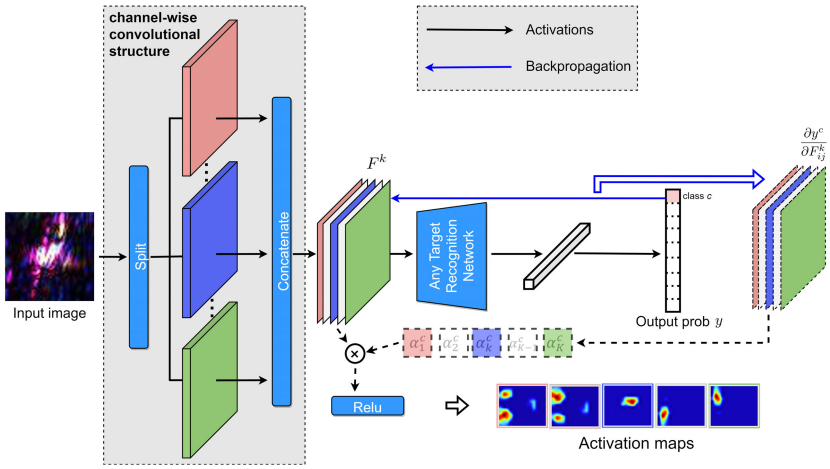


Fig. 2. Proposed contribution visualization approach overview: Given an image and a class of interest (class c), we forward propagate the image through the proposed channel-wise convolutional structure and then through any target recognition network to obtain a raw score (output probability) for the category. This signal is then backpropagated to the feature maps of interest ($F^k, k = 1, 2, \ldots, K$), which we combine to compute the importance weight $\alpha^c_k, k = 1, 2, \ldots, K$, and finally compute the activation maps which represents the attention of the target recognition network model. Since the feature map F^k is unique to the kth input channel, the activation maps reflect the contribution and importance of the input channel.

we introduce a unique and simple polarimetric feature, capable of adequately characterizing the physical scattering structure of the target.

A. Polarimetric SAR Data

For single-look polarimetric SAR data, each resolution cell (pixel) is represented by a 2×2 Sinclair matrix (also known as polarimetric scattering matrix). In the linear horizontal and vertical polarization base, this matrix can be expressed as

$$\mathbf{S} = \begin{bmatrix} S_{hh} & S_{hv} \\ S_{vh} & S_{vv} \end{bmatrix} \tag{3}$$

where S_{kl} stands for the scattering coefficient of k-transmitting and l-receiving polarizations. When the reciprocity holds, the scattering matrix is symmetrical, that is, $S_{hv} = S_{vh}$. Then, in the Lexicographic basis, the matrix $\mathbf{S}$ can be vectorized as

$$\boldsymbol{k}_L = \begin{bmatrix} S_{hh} & \sqrt{2}S_{hv} & S_{vv} \end{bmatrix}^T \tag{4}$$

where the superscript T denotes the matrix transpose. In the multilook case, each pixel is represented by the polarimetric covariance matrix $\mathbf{C}$

$$\begin{aligned} \mathbf{C} &= \left\langle \boldsymbol{k}_L \boldsymbol{k}_L^H \right\rangle \\ &= \begin{bmatrix} \left\langle |S_{hh}|^2 \right\rangle & \sqrt{2}S_{hh}S_{hv}^* & S_{hh}S_{vv}^* \\ \sqrt{2}S_{hv}S_{hh}^* & \left\langle 2|S_{hv}|^2 \right\rangle & \sqrt{2}S_{hv}S_{vv}^* \\ S_{vv}S_{hh}^* & \sqrt{2}S_{vv}S_{hv}^* & \left\langle |S_{vv}|^2 \right\rangle \end{bmatrix} \end{aligned} \tag{5}$$

where $\langle \cdot \rangle$ denotes the ensemble average, the superscript H denotes the conjugate transpose, and the superscript $*$ stands for the complex conjugate.

B. Contribution Visualization Approach

To illuminate the significance of polarimetric information in target recognition, we introduce a contribution visualization approach by tailoring Grad-CAM to our proposed channel-wise convolutional structure. A comprehensive overview of the approach is depicted in Fig. 2. Specifically, given an input image $X_{\text{input}} \in \mathbb{R}^{h \times w \times K}$, we propose a channel-wise convolutional structure H_{cwcs} to generate shallow feature maps

$$F = H_{\text{cwcs}}\left(X_{\text{input}}\right). \tag{6}$$

The number of feature maps corresponds to the number of input channels. Assuming an image input with K channels, the proposed channel-wise convolutional structure comprises K channel-wise convolutional layers. We have

$$\begin{aligned} F &= H_{\text{cwcs}}(X_{\text{input}}) \\ &= \text{concat}\Big(\text{conv}_1(X^1_{\text{input}}, 1, 1), \ldots, \\ &\qquad \text{conv}_k(X^k_{\text{input}}, 1, 1), \ldots, \text{conv}_K(X^K_{\text{input}}, 1, 1)\Big) \end{aligned} \tag{7}$$

where $\text{conv}_k(X^k_{\text{input}}, 1, 1) \in \mathbb{R}^{h \times w \times 1}$ denotes the proposed kth channel-wise convolutional layer whose input channel number and output channel number are both 1, and X^k_{input} is the kth channel of the image X_{input}, and concat represents the concatenation operation that concatenate the outputs of the K channel-wise convolutional layers in channel direction. The feature maps $F \in \mathbb{R}^{h \times w \times K}$ are then used for further processing by any target recognition network H_{network}. We have

$$y = H_{\text{network}}(F) \tag{8}$$

where $y \in \mathbb{R}^C$ denotes the predicted probabilities for C classes of the proposed networks.

As discussed in Section II-C, the Grad-CAM [48] serves as a tool to interpret potential network decisions, generating the activation map by calculating gradient information between the final convolutional layer and the class score. Our objective is to employ Grad-CAM to discern the contribution of individual polarization channel elements within the covariance matrix. However, the limitation lies in Grad-CAM's inability to visualize the input contribution directly. The proposed solution to this complex issue involves using our

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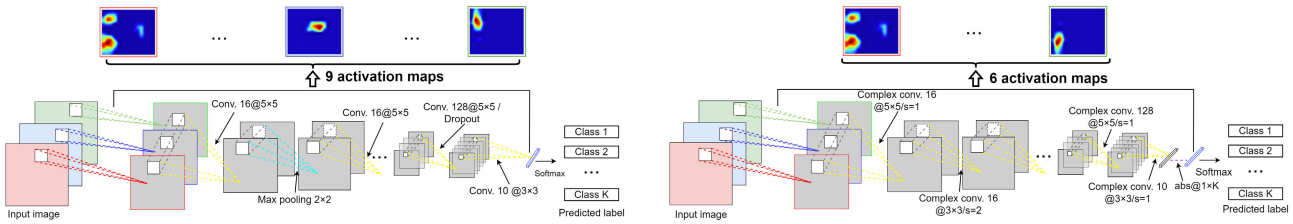


Fig. 3. Overall architecture of the proposed (a) cvA-ConvNets and (b) cvCV-FCNN. The yellow dotted lines denote the convolutional layers, and the cyan dotted ones denote the max-pooling layers. The convolution layers are annotated by "Conv. (number of feature maps) @ (filter size)" or "Complex conv. (number of feature maps) @ (filter size)/s=(stride step size)," and the pooling layers are annotated by "Max pooling (pooling size)." The proposed channel-wise convolutional structure is represented by the red, blue, and green dotted lines. The number of activation maps depends on the number of channels in the proposed channel-wise structure.

channel-wise convolutional structure as a proxy for the input, thereby enabling the adaptation of Grad-CAM to visualize the activation of the channel-wise convolutional layers. Consequently, the contribution of polarimetric information becomes identifiable and accentuated. Specifically, we compute the gradient if the score of class c, y^c (before the softmax), with respect to the kth feature map F^k generated by the kth channel-wise convolutional layer conv_k, and then obtain the neuron importance weight α_k^c via global average pooling

$$\alpha_k^c = \frac{1}{Z}\sum_i \sum_j \frac{\partial y^c}{\partial F_{ij}^k} \tag{9}$$

where

$$F^k = \text{conv}_k(X_{\text{input}}^k, 1, 1). \tag{10}$$

The final class-discriminative activation map $L_k^c \in \mathbb{R}^{h\times w}$ for any class c with respect to the kth channel is defined as

$$L_k^c = \text{norm}_c\left(\text{ReLU}\left(\alpha_k^c F^k\right)\right) \tag{11}$$

where norm_c denotes the channel-wise normalization. The activation map L_k^c signifies the influence of the feature map F^k on the classification score for class c. It is worth noting that feature map F^k relates solely to the input of the kth channel in our proposed channel-wise convolutional structure. Hence, the activation map inherently demonstrates the contribution of the kth channel input to the classification score.

The proposed visualization approach is based on Grad-CAM, but there are significant differences. First, the original Grad-CAM always calculates the activation maps of the last convolutional feature map with respect to the score of some class, while our approach focusses on the contributions of middle layers to the final results. Then, the authors of Grad-CAM perform a weighted combination of forward maps to generate a single comprehensive map. Our visualization method not only highlights the important regions in the image (which the original Grad-CAM does), but also points out the polarimetric channels of more interest. Finally, Grad-CAM is unable to visualize the input contribution directly, while our method avoids the problem by using our channel-wise convolutional structure as a proxy for the input.

C. cvA-ConvNets and cvCV-FCNN

To achieve more comprehensive performance evaluation and analysis, real- and complex- form data are used in conjunction with both network structures. We develop cvA-ConvNets and cvCV-FCNN by applying the proposed contribution visualization approach to A-ConvNets and CV-FCNN. Their overall architectures are illustrated in Fig. 3. In Fig. 3(a), the convolutional layers are represented by the yellow dotted lines and annotated by "Conv. (number of feature maps) @ (filter size);" the max-pooling layers are represented by the cyan dotted lines and annotated by "Max pooling (pooling size)." In Fig. 3(b), the convolutional layers refer to the complex-valued convolutional layers, which are also represented by the yellow dotted lines and annotated by "Complex conv. (number of feature maps) @ (filter size)/s=(stride step size)." The proposed channel-wise convolutional structures are represented by the red, blue, and green dotted lines.

The cvA-ConvNets utilizes real-valued convolutional layers, while the cvCV-FCNN employs complex-valued convolutional layers. We vectorize the polarimetric covariance matrix $\mathbf{C}$ in the real-valued form as

$$\begin{aligned}\text{vec_real}(\mathbf{C}) = &\left[\left\langle|S_{hh}|^2\right\rangle, \left\langle 2|S_{hv}|^2\right\rangle, \left\langle|S_{vv}|^2\right\rangle\right.\\ &\text{real}\left(\sqrt{2}S_{hh}S_{hv}^*\right), \text{real}\left(S_{hh}S_{vv}^*\right)\\ &\text{real}\left(\sqrt{2}S_{hv}S_{vv}^*\right), \text{imag}\left(\sqrt{2}S_{hh}S_{hv}^*\right)\\ &\left.\text{imag}\left(S_{hh}S_{vv}^*\right), \text{imag}\left(\sqrt{2}S_{hv}S_{vv}^*\right)\right]^T\end{aligned} \tag{12}$$

for cvA-ConvNets, where real() and imag() represent the operation of taking the real part and imaginary part of a complex number, respectively. In this case, the input image is defined as $X_{\text{input}} \in \mathbb{R}^{h\times w\times 9}$. For cvCV-FCNN, the polarimetric covariance matrix $\mathbf{C}$ is vectorized in the complex-valued form as

$$\begin{aligned}\text{vec_complex}(\mathbf{C}) = &\left[\left\langle|S_{hh}|^2\right\rangle, \left\langle 2|S_{hv}|^2\right\rangle, \left\langle|S_{vv}|^2\right\rangle\right.\\ &\left.\sqrt{2}S_{hh}S_{hv}^*, S_{hh}S_{vv}^*, \sqrt{2}S_{hv}S_{vv}^*\right]^T\end{aligned} \tag{13}$$

and the input image X_{input} has a shape of $h \times w \times 6$.

D. Polarimetric Feature-Based Target Recognition

1) Polarimetric Correlation Feature: The quantitative analysis of polarimetric information in the experimental sections (Section IV-C and IV-D) reveals that the correlation between co-polarization and cross-polarization bolsters SAR target recognition. Inspired by these conclusions, we propose a

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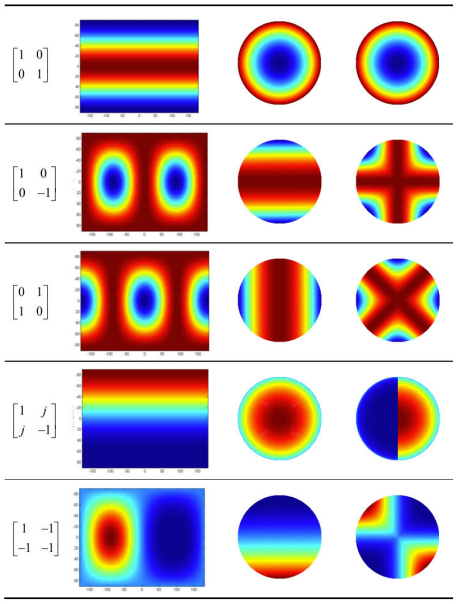


Fig. 4. Typical scatterers and their co-polarization signature. From top to bottom: the surface scatterer, diplane scatterer, diplane scatterer with a relative orientation of 45°, dipole scatterer, and helix scatterer.

simple polarimetric correlation feature for target recognition, which is defined as the correlation between the target and typical scatterers. Common typical scatterers include the surface scatterer, diplane scatterer, diplane scatterer with a relative orientation of 45°, dipole scatterer, and helix scatterer, whose scattering matrices and co-polarization signatures are shown in Fig. 4. Generally speaking, the conventional selection is the first three typical scatterers, which constitute the classic Pauli spin matrix basis set

$$\boldsymbol{\Psi}_P = \left\{ \sqrt{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \sqrt{2}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \sqrt{2}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}. \tag{14}$$

However, the second and third scatterers can be regarded as the result of the same scatterer at different orientation angles (0° and 45°). In Fig. 4, we can also see that the signatures of the second and the third only differ by a rotation angle or translation. Hence, in this article, we select the following typical scatterers for correlation feature definition: the surface, the diplane, the dipole, and the helix. The dipole scatterer refers to structures such as the barrel on the tank target and gables on the vehicle target, and the helix scatterer represents the asymmetric scatterers on the target. In the subsequent experimental section, we will show that such a typical scatterer selection is superior to other options, including the conventional selection.

The vectorized representation of the scattering matrices for the four selected scatterers can be expressed as

$$\begin{aligned} \boldsymbol{k}_{L,s} &= \left[1/\sqrt{2},\ 0,\ 1/\sqrt{2}\right]^T \\ \boldsymbol{k}_{L,\text{dipl}} &= \left[1/\sqrt{2},\ 0,\ -1/\sqrt{2}\right]^T \\ \boldsymbol{k}_{L,\text{dipo}} &= \left[1/2,\ -\sqrt{2}/2,\ -1/2\right]^T \\ \boldsymbol{k}_{L,h} &= \left[1/2,\ \sqrt{2}j/2,\ -1/2\right]^T. \end{aligned} \tag{15}$$

The proposed polarimetric feature is defined as a correlation between the target and another scatterer. For single-look polarimetric data, denote by $\mathbf{S}_1$ and $\mathbf{S}_2$ the target scattering matrix and the other scattering matrix, respectively. We have

$$r_s\left(\boldsymbol{k}_{L,1}, \boldsymbol{k}_{L,2}\right) = \left|\boldsymbol{k}_{L,2}^H \boldsymbol{k}_{L,1}\right|^2 \tag{16}$$

where $\boldsymbol{k}_{L,i}$ is the vectorized representation of $\mathbf{S}_i$ $(i = 1, 2)$. Applying (16) to the four typical scatterers, we obtain the correlation features with respect to the surface, the diplane, the dipole, and the helix

$$\begin{aligned} r_{\text{surface}} &= r_s\left(\boldsymbol{k}_L, \boldsymbol{k}_{L,s}\right) \\ r_{\text{diplane}} &= r_s\left(\boldsymbol{k}_L, \boldsymbol{k}_{L,\text{dipl}}\right) \\ r_{\text{dipole}} &= r_s\left(\boldsymbol{k}_L, \boldsymbol{k}_{L,\text{dipo}}\right) \\ r_{\text{helix}} &= r_s\left(\boldsymbol{k}_L, \boldsymbol{k}_{L,h}\right). \end{aligned} \tag{17}$$

For multilook polarimetric data, the correlation between the covariance matrix $\mathbf{C}_1$ and covariance matrix $\mathbf{C}_2$ is defined as

$$r_c(\mathbf{C}_1, \mathbf{C}_2) = \left|\text{Tr}\left(\mathbf{C}_2^H \mathbf{C}_1\right)\right| \tag{18}$$

where $\text{Tr}(\cdot)$ denotes taking the trace of a matrix. Correspondingly, the correlation features with respect to the four scatterers for multilook data can be expressed as

$$\begin{aligned} r_{\text{surface}} &= r_c\left(\left\langle \boldsymbol{k}_L \boldsymbol{k}_L^H \right\rangle, \boldsymbol{k}_{L,s} \boldsymbol{k}_{L,s}^H\right) \\ r_{\text{diplane}} &= r_c\left(\left\langle \boldsymbol{k}_L \boldsymbol{k}_L^H \right\rangle, \boldsymbol{k}_{L,\text{dipl}} \boldsymbol{k}_{L,\text{dipl}}^H\right) \\ r_{\text{dipole}} &= r_c\left(\left\langle \boldsymbol{k}_L \boldsymbol{k}_L^H \right\rangle, \boldsymbol{k}_{L,\text{dipo}} \boldsymbol{k}_{L,\text{dipo}}^H\right) \\ r_{\text{helix}} &= r_c\left(\left\langle \boldsymbol{k}_L \boldsymbol{k}_L^H \right\rangle, \boldsymbol{k}_{L,h} \boldsymbol{k}_{L,h}^H\right). \end{aligned} \tag{19}$$

The final polarimetric correlation feature is given by a concentration of the four correlations defined above

$$\boldsymbol{x}_{\text{corr}} = \left[r_{\text{surface}},\ r_{\text{diplane}},\ r_{\text{diplane45°}},\ r_{\text{helix}}\right]^T. \tag{20}$$

2) Relationship to Similarity Parameters: The similarity parameter for single-look SAR data proposed by Yang et al. [38] takes the form of

$$p_s\left(\boldsymbol{k}_{L,1}, \boldsymbol{k}_{L,2}\right) = \frac{\left|\boldsymbol{k}_{L,1}^H \boldsymbol{k}_{L,2}\right|^2}{||\boldsymbol{k}_{L,1}||_2^2 ||\boldsymbol{k}_{L,2}||_2^2}. \tag{21}$$

For the four typical scatterers, the modulus of the vectorized representation is 1, namely

$$\left\|\boldsymbol{k}_{L,i}\right\|_2^2 = 1, i \in \{s, \text{dipl}, \text{dipo}, h\}. \tag{22}$$

We can note that

$$r_s\left(\boldsymbol{k}_L, \boldsymbol{k}_{L,i}\right) = p_s\left(\boldsymbol{k}_L, \boldsymbol{k}_{L,i}\right) ||\boldsymbol{k}_L||_2^2 \tag{23}$$

where $i \in \{s, \text{dipl}, \text{dipo}, h\}$. The modulus of the vectorized representation $\boldsymbol{k}_L$ can be easily interpreted as the intensity of the target. That is, the proposed correlation feature amalgamates intensity information and similarity measures [38], [39] associated with the four typical scatterers. The same is true for multilook polarimetric SAR data.

Intensity can delineate strong scatter points of a target and the corresponding geometric patterns, particularly under high SCR conditions. However, the acquisition of more nuanced physical structures remains elusive when relying solely on

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intensity information. Furthermore, as the SCR diminishes, the performance boost conferred by the intensity information progressively declines to nil. Hence, we suggest employing polarimetric correlation features to portray the physical structures of the target. The position and dimensions of typical physical structures such as surface scatterers and diplane scatterers vary substantially across different target classes. By incorporating intensity and correlation features associated with surface scatterers, diplane scatterers, and helix scatterers, we attain a thorough and comprehensive characterization of the target's physical structure.

The polarimetric correlation feature proposed in this article has the following significant differences with the similarity parameter. The similarity parameter does not contain intensity information, and the target is easily submerged in the background; while our feature combines intensity information and target similarity, highlighting the target while depicting the component structure of the target. In addition, four typical scatterers are specified to generate a comprehensive description of the target, discarding the commonly used Pauli basis-based scatterer set. We will demonstrate the effectiveness of the typical scatterer selection through subsequent ablation experiments.

3) Target Recognition Scheme: Given a polarimetric SAR image with the size of $h \times w$, we extract the polarimetric correlation feature $X_{\text{corr}} \in \mathbb{R}^{h\times w\times 4}$ according to (20). Then, the extracted feature X_{corr} is input into the conventional A-ConvNets [26] to generate the final predicted classification label.

The A-ConvNets only include five convolutional layers as well as some activation and pooling layers. We use it as a backbone precisely because of the simplicity of the network. Our purpose is to verify the effectiveness of the proposed features, so the network is set up as simply as possible. If a carefully designed deeper network is used, the recognition performance of the proposed features will be further improved.

IV. Experiments and Discussions

A. Experimental Datasets

1) Pol-MSTAR: The Pol-MSTAR dataset, a synthetic creation of the State Key Laboratory of Millimeter Waves at Southeast University, builds on the standard MSTAR dataset and a geometric physical model of military targets. Sophisticated numerical calculations of electromagnetic scattering generate the fully polarimetric responses included in this dataset. However, due to the lack of an appropriate physical model, the D7 class from the standard MSTAR dataset is absent from Pol-MSTAR. Fig. 5 illustrates the types of military targets represented in the Pol-MSTAR dataset. The top row depicts the optical image, while the bottom row visualizes polarimetric SAR data using the Pauli basis.

2) GOTCHA: The GOTCHA dataset [49], collected by AFRL, is a fully polarimetric dataset consisting of eight complete circular passes (covering a full azimuthal rotation of 360°) at varying depression angles. The scene image includes numerous calibration targets and civilian ground vehicles. Fig. 6 displays the scene of interest, visualized using the Pauli basis, where the nine vehicle targets are delineated by green boxes and labeled with green numbers. The corresponding optical images of these nine vehicle categories are presented in Fig. 7.

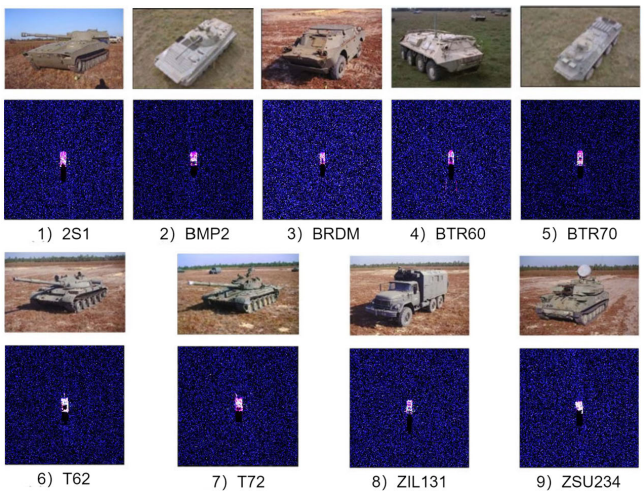


Fig. 5. Types of military targets in Pol-MSTAR: Optical images (top) juxtaposed with visualization images of polarimetric SAR data utilizing the Pauli basis (bottom). Note: D7 from the standard MSTAR dataset is not included in Pol-MSTAR.

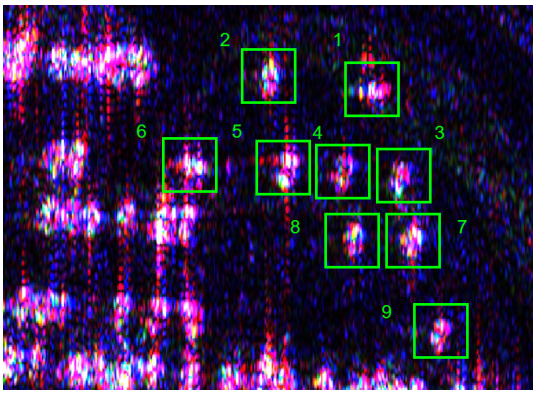


Fig. 6. Scene image with the Pauli basis in GOTCHA. The nine-class vehicle targets are marked by green boxes and labeled by green numbers.



Fig. 7. Optical images of the nine-class vehicle targets in GOTCHA.

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TABLE I
DETAILS OF THE TRAINING AND TEST SETS IN POL-MSTAR AND GOTCHA

Dataset	Split	Depression	Image size	Number
Pol-MSTAR	Training set	17°	90×90	360×9
	Test set	15°	90×90	360×9
GOTCHA	Training set	$[43.7°, 45°]$	54×54	$4 \times 360 \times 9$
	Test set	$[43.7°, 45°]$	54×54	$4 \times 360 \times 9$

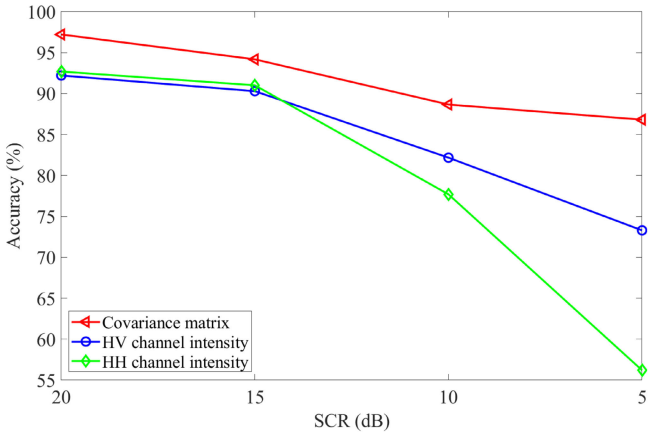


Fig. 8. Classification accuracy with respect to SCR, using different types of inputs with cvA-ConvNets on synthetic Pol-MSTARs. As SCR continually diminishes, the classification accuracy of the HH channel intensity undergoes the most drastic reduction, followed by the HV intensity and finally the covariance matrix.

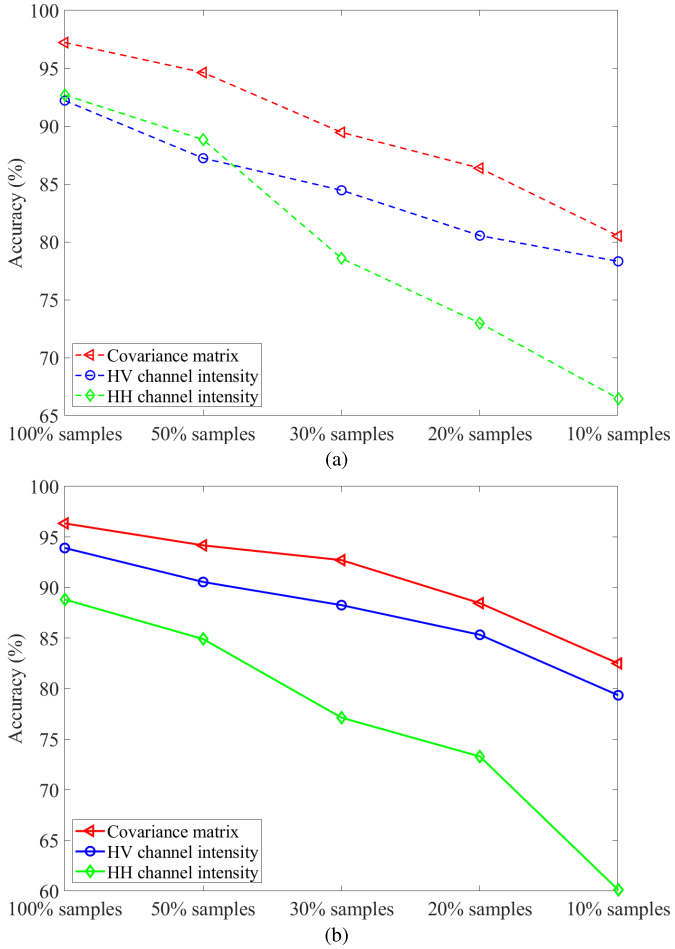


Fig. 9. Classification accuracy with respect to the sizes of training sets on (a) Pol-MSTAR and (b) GOTCHA, using different types of inputs with cvA-ConvNets. Utilizing the covariance matrix or HV channel intensity exhibits substantial superiority over using the HH channel intensity with a limited number of training samples, which indicates that polarimetric information enables the network to learn more robust and discriminative feature representation.

B. Implementation Details and Evaluation Metrics

The specifications for the training and test sets within the Pol-MSTAR and GOTCHA datasets are presented in Table I. Image samples in the Pol-MSTAR dataset are sized at 90×90 pixels, while those in the GOTCHA dataset are 54×54 pixels. For Pol-MSTAR, the training images are taken at a depression angle of 17°, while the test images are captured at 15°. Each set, both training and testing, includes 360 samples per class, encompassing a total of nine classes. In the GOTCHA dataset, each pass is designed with an ideal elevation separation of $\Delta\theta = 0.18°$, within a planned elevation range of $\theta \in [43.7°, 45°]$. Images from passes 1, 3, 5, and 7 are selected for training, while images from passes 2, 4, 6, and 8 are allocated for testing. Consequently, the size of the training and testing sets stands at $4 \times 360 \times 9$.

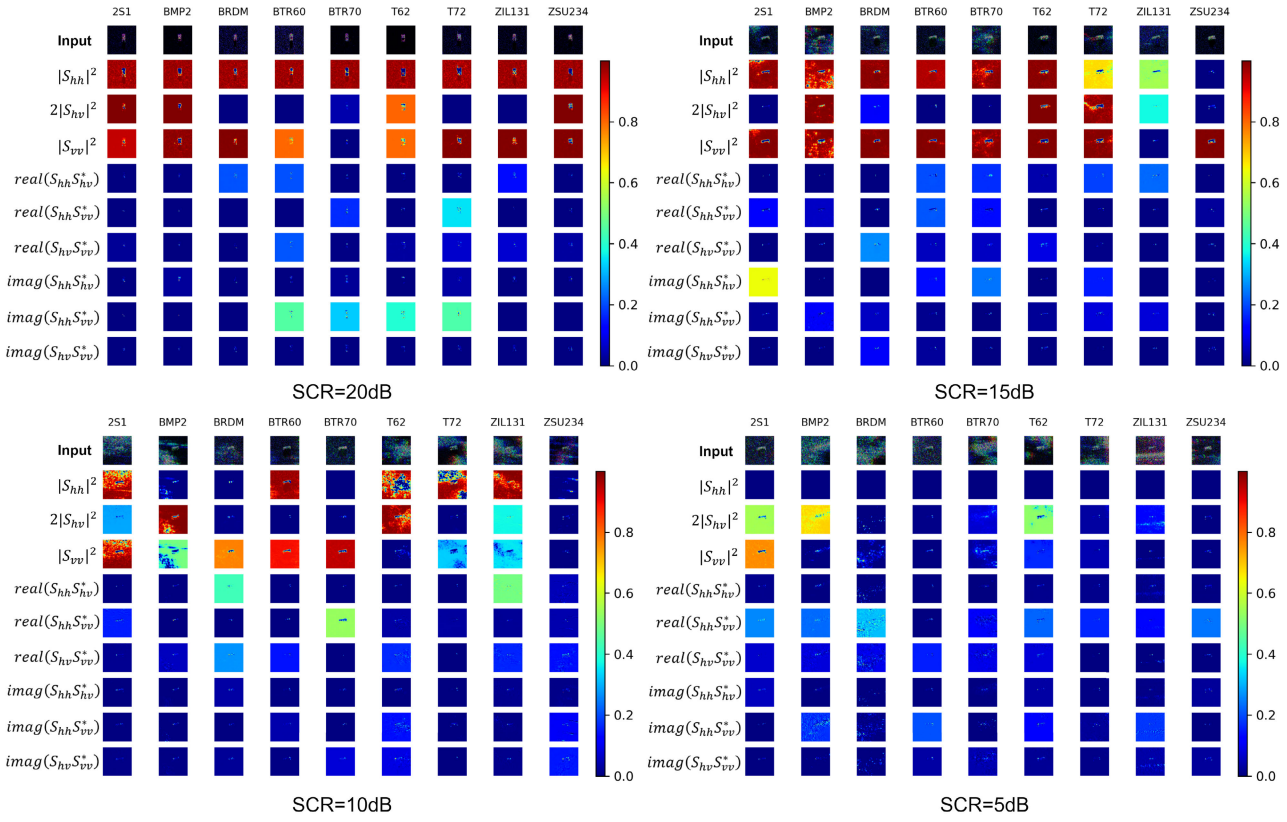
Regarding the training process, the weight parameters of the proposed networks are initialized using Kaiming initialization. The batch size is fixed at 8, and an initial learning rate of 4e-4 is established, which gradually decreases according to a Cosine schedule. The networks are trained over 300 epochs. Data augmentation procedures employed include random cropping and clipping. Notably, 80% of the training set is allocated for training, with the remaining portion set aside for validation.

Referring to relevant SAR target recognition works [26], [27], [28], [47], we use three metrics including the overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa), as well as the confusion matrix for performance evaluation. OA is defined as the ratio of the number of correctly classified test samples to the total number of test samples, while AA is defined as the AA of each class, where the accuracy of each class is defined as the ratio of the number of correctly classified samples of each class to the total number of test samples of that class. Kappa combines the diagonal and off-diagonal terms of the confusion matrix and is a robust consistency measure. Notably, since the number of samples in each category of the two datasets is the same, OA and AA are also the same. Unless otherwise specified, the accuracy in the following figures and tables below refers to OA or AA.

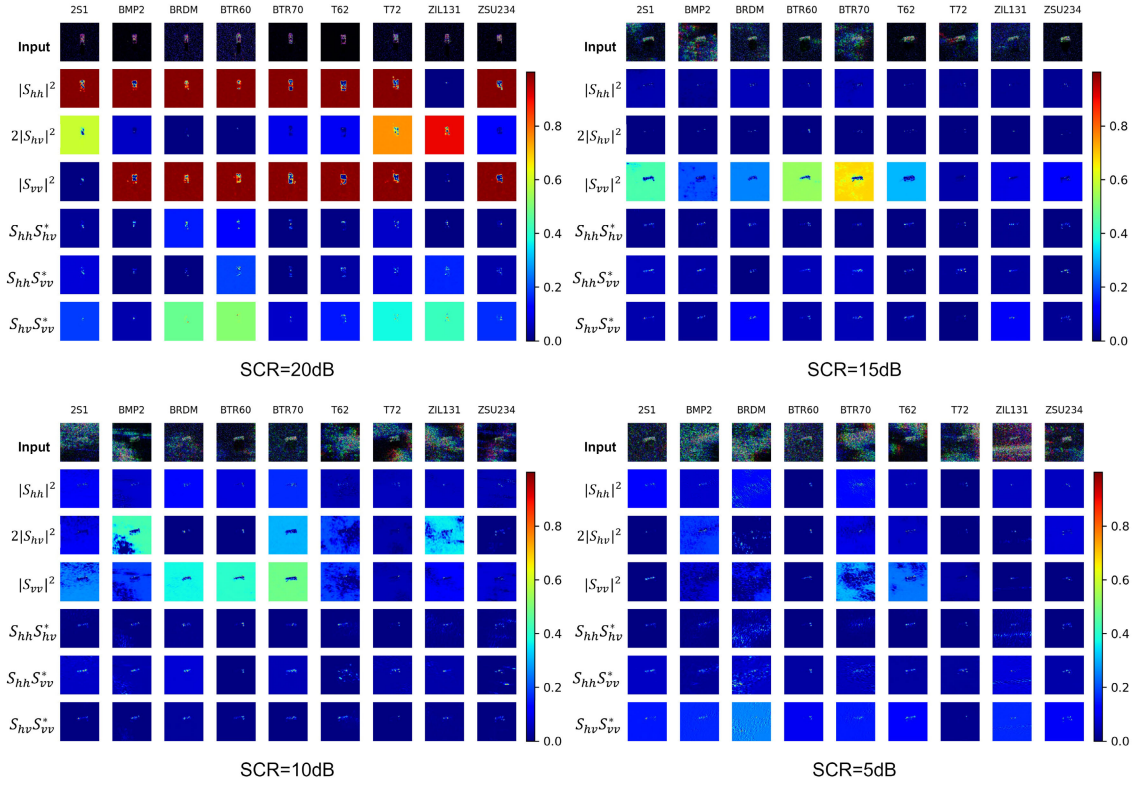
C. Advantages of Polarimetric SAR Data

To assess the impact of clutter on recognition performance, we generate target backgrounds of varying intensities using scene data, which maintains the same frequency, resolution, and other parameters as the Pol-MSTAR dataset. Background images are randomly selected to synthesize new images with differing SCRs. Fig. 8 depicts the classification accuracy in relation to SCR, employing different types of inputs with cvA-ConvNets on synthetic Pol-MSTAR images.

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(a)



(b)

Fig. 10. Activation maps of (a) cvA-ConvNets and (b) cvCV-FCNN on Pol-MSTAR with different SCRs. The activation maps are arranged by channel types in rows and by classes in columns. As SCR continues to decrease, the weight of the network on the channel of the diagonal elements gradually decreases, the weight of the elements on the off-diagonal gradually increases, and the focus of the network, namely the high-weight channel, shows a shift from the diagonal to the off-diagonal channels.

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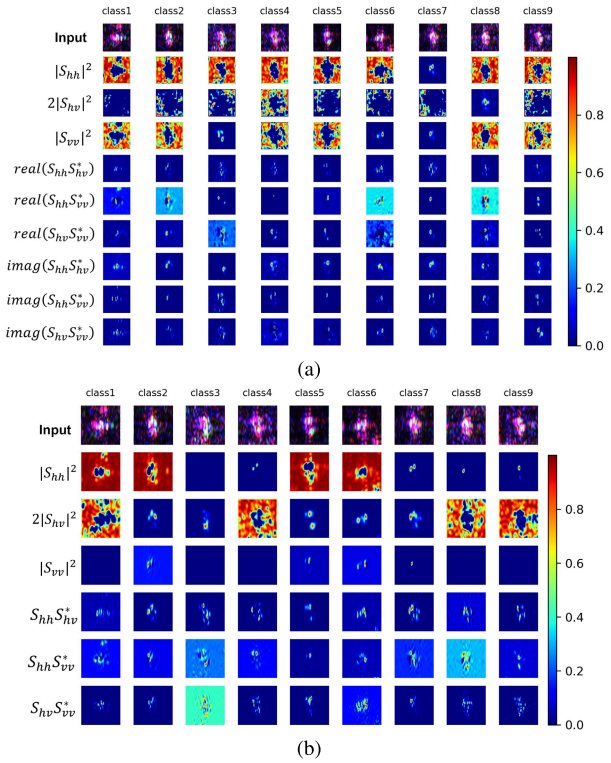


Fig. 11. Activation maps of (a) cvA-ConvNets and (b) cvCV-FCNN on GOTCHA. In each subfigure, the activation maps are arranged by channel types in row direction, and by classes in column direction. HV intensity channel ($2|S_{hv}|^2$) and correlation channel between cross-polarization and co-polarization ($S_{hh}S_{hv}^*$ and $S_{hv}S_{vv}^*$) maintain higher weights and show higher contrast between the target and the background.

TABLE II
TARGET RECOGNITION ACCURACY (%) COMPARISON WITH CVA-CONVNETS ON POL-MSTAR AND GOTCHA

	Input types	Pol-MSTAR	GOTCHA
Diagonal elements	$\|S_{hh}\|^2$	77.69	90.82
	$2\|S_{hv}\|^2$	82.17	95.91
	$\|S_{vv}\|^2$	79.70	88.62
Off-diagonal elements	$S_{hh}S_{hv}^*$	82.53	95.25
	$S_{hh}S_{vv}^*$	78.87	93.85
	$S_{hv}S_{vv}^*$	83.89	95.41

As anticipated, with the continual decrease in SCR, the classification accuracy suffers the most when using the HH channel intensity, followed by the HV intensity, while the accuracy decrement is least pronounced when using the covariance matrix.

Intensity information most directly reflects the physical and structural attributes of the target. Due to its simplicity, such information is readily captured and learned by the network. Polarimetric information, however, encapsulates both intensity and phase. Even in instances where intensity information proves insufficient, target recognition can still be accomplished based on phase information. In SAR target recognition tasks, polarimetric information can effectively suppress the interference of clutter, outperforming single-polarization intensity information.

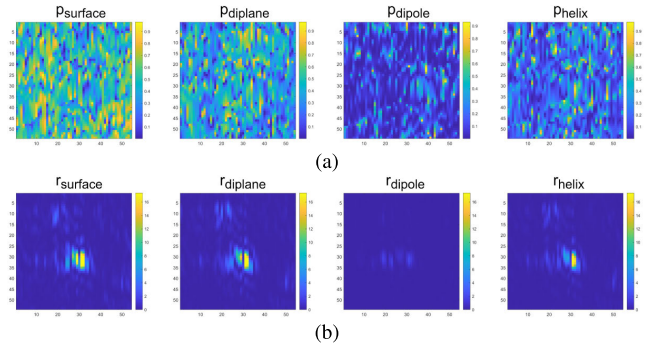


Fig. 12. Visualization comparison between (a) normalized similarity parameters and (b) proposed correlation features. Evidently, the correlation features more distinctly highlight the target against the background, compared to the normalized similarity parameters.

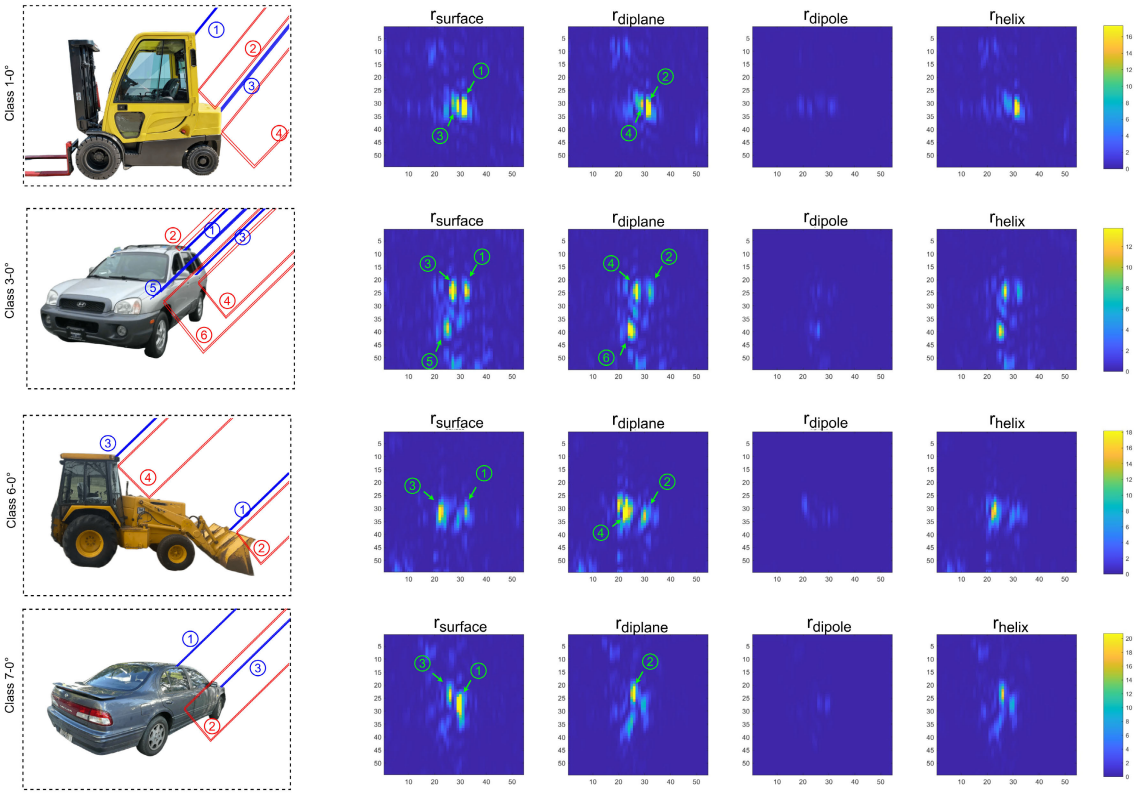
Fig. 9 illustrates the classification accuracy in relation to the sizes of the training sets, using various types of inputs with cvA-ConvNets. We can see that the use of a covariance matrix or HV channel intensity demonstrates significant advantages over the use of HH channel intensity, particularly when the quantity of training samples is limited. This suggests that polarimetric information enables the network to learn more robust and discriminative feature representations, thereby ensuring that training with a small number of samples can still yield satisfactory recognition performances.

D. Contributions of the Polarimetric Covariance Matrix Elements

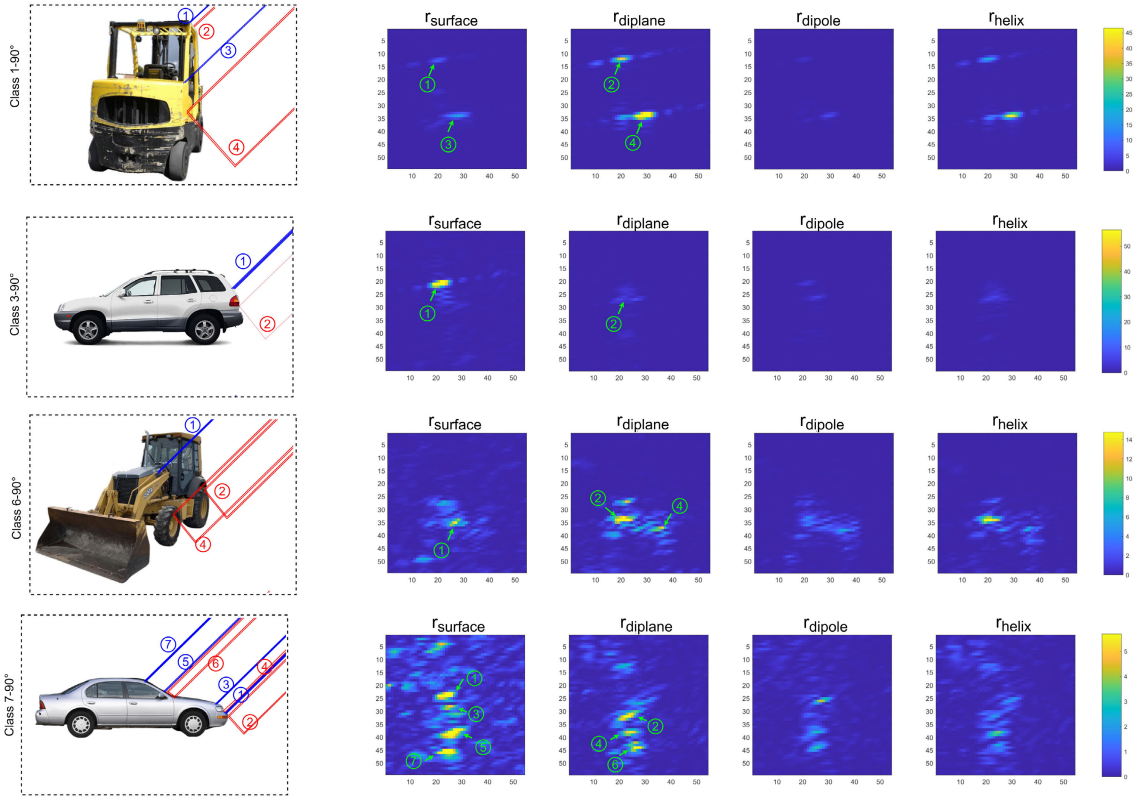
Fig. 10 displays the activation maps of (a) cvA-ConvNets and (b) cvCV-FCNN on the synthetic Pol-MSTAR images. The activation maps are organized by channel types row-wise and by classes column-wise. When the SCR is high, the network weight on the diagonal element channel of the polarimetric covariance matrix significantly surpasses other channel weights. As the SCR diminishes, the weight of the network on the diagonal elements' channel gradually decreases, and the weight on the off-diagonal elements progressively increases. Consequently, the network's focus—represented by the high-weight channel—shifts from the diagonal to the off-diagonal elements. These results further corroborate our Section IV-C conclusion that while the network easily learns intensity information due to its simplicity, it falls short in low SCR conditions. Conversely, polarimetric information, not solely dependent on intensity, remains pertinent even under low SCR circumstances.

In Fig. 10, the diagonal element $|S_{vv}|^2$ consistently maintains a high weight. Among the off-diagonal elements, $S_{hv}S_{vv}^*$ exhibits relatively high weights. Channels related to VV also demonstrate relatively high weights, possibly due to the disparity between the simulation and actual data on the VV channel. Fig. 11 presents the activation maps of cvA-ConvNets and cvCV-FCNN on the real image dataset GOTCHA. Among the diagonal elements, $|S_{hh}|^2$ and $|2S_{hv}|^2$ outweigh $|S_{vv}|^2$. Among the off-diagonal elements, $S_{hh}S_{vv}^*$ carries higher weights than $S_{hh}S_{hv}^*$ and $S_{hv}S_{vv}^*$. However, the activation maps of the $S_{hh}S_{vv}^*$ channel exhibit a lower contrast between target and background, indicating that the network has not effectively

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(a)



(b)

Fig. 13. Mapping the target physical geometry to the polarimetric features at (a) 0° azimuth and (b) 90° azimuth. Four typical targets in GOTCHA are presented. From top to bottom: Hyster Forklift, Hyundai Santa Fe, CASE Tractor, and Nissan Maxima. Odd-bounce scatterings are represented by blue lines and marked by odd numbers, while even-bounce scatterings are represented by red lines and marked by even numbers. The positions on the feature component maps corresponding to the geometric structures are marked by green numbers.

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TABLE III
CLASSIFICATION ACCURACY (%) ON POL-MSTARS AND GOTCHA. THE BEST RESULTS ARE EMPHASIZED IN BOLDFACE

Class	Pol-MSTAR 20dB				Pol-MSTAR 10dB				GOTCHA			
	Proposed	Y4O-based	CV-3D-CNN	MS-CVNN	Proposed	Y4O-based	CV-3D-CNN	MS-CVNN	Proposed	Y4O-based	CV-3D-CNN	MS-CVNN
1	97.22	94.29	96.18	96.72	81.16	77.84	73.18	76.50	99.86	95.86	95.06	95.33
2	94.18	89.27	94.24	93.41	75.34	78.94	71.79	76.22	97.36	90.03	93.53	93.32
3	98.61	95.67	98.11	98.11	85.60	77.28	82.59	81.49	99.24	94.12	94.99	94.92
4	92.79	92.91	93.13	92.63	78.11	78.39	74.29	74.01	98.61	90.86	95.19	94.78
5	96.68	94.84	97.28	96.45	87.81	85.87	81.76	81.77	98.47	91.83	91.86	93.25
6	100	98.17	100	100	92.24	88.36	88.97	83.03	99.93	96.00	96.58	96.44
7	95.84	94.84	95.95	95.90	86.70	80.05	81.21	78.99	99.90	94.68	95.19	95.12
8	96.95	96.45	93.40	96.45	84.76	78.39	78.63	79.27	99.65	95.44	95.68	93.68
9	96.68	95.12	93.95	96.45	85.04	75.07	79.83	71.24	96.60	90.19	92.49	94.57
OA/AA	**96.55**	94.26	95.80	96.23	**84.09**	80.02	79.08	78.50	**98.78**	93.22	94.50	94.82
Kappa	**96.03**	93.15	94.62	95.97	**81.98**	77.98	76.84	74.22	**98.51**	92.09	93.67	93.86

extracted the target structure on this channel. This suggests that the correlations between cross-polarization and co-polarization ($S_{hh}S_{hv}^*$ and $S_{hv}S_{vv}^*$) are crucial components.

To analyze the contributions of the polarimetric covariance matrix, we select a single matrix element or their combination for network input during training. By comparing the classification accuracy on the test set, we can quantitatively evaluate the matrix elements' contributions. Table II presents the classification accuracy on Pol-MSTAR and GOTCHA using matrix elements as input. On the Pol-MSTAR dataset, the HV channel intensity ($2|S_{hv}|^2$) performs optimally among the diagonal elements. The correlation between cross-polarization and co-polarization ($S_{hh}S_{hv}^*$ and $S_{hv}S_{vv}^*$) outperforms the correlation between co-polarizations ($S_{hh}S_{vv}^*$). Experimental results on GOTCHA are consistent with Pol-MSTAR results.

In conclusion, among the elements of the polarimetric covariance matrix, the cross-polarization channel's intensity and the correlation between cross-polarization and co-polarization emerge as key components.

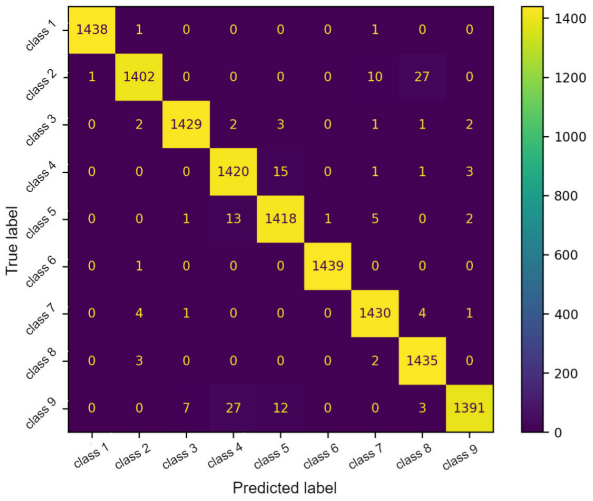


Fig. 14. Confusion matrix of the proposed polarimetric feature on GOTCHA.

E. Performance of the Proposed Polarimetric Feature

1) Mapping to Target Physical Geometry: We begin by computing both normalized and unnormalized similarity parameters, comparing the resulting visualizations in Fig. 12. Targets are easily distinguishable from the background in the unnormalized parameter visualizations, while in the normalized parameter visualizations, the target and the background tend to blend together. Thanks to the combination of intensity and polarimetric information, the unnormalized parameters manage to differentiate the target from the background effectively.

We carry out a meticulous analysis of the proposed polarimetric feature and establish a mapping relationship between the feature components and the target's physical geometry, as displayed in Fig. 13. This figure includes four typical targets from the GOTCHA dataset, listed from top to bottom as Hyster Forklift, Hyundai Santa Fe, CASE Tractor, and Nissan Maxima. Given the difficulty of finding a target picture identical to the vehicle used in the GOTCHA data collection experiment by the AFRL, we select a target picture with the same model specifications and physical geometric structure as the vehicle used in the collection experiment. The targets depicted in the leftmost diagrams in Fig. 13(a) and (b) differ insignificantly from the real targets, with minor variations such as color. We represent odd-bounce scatterings with blue lines, marking them with odd numbers, and depict even-bounce scatterings with red lines, marked by even numbers.

As we can observe, the appearance of the same target differs substantially in SAR images from different azimuth angles due to the SAR imaging working mode. More importantly, despite these disparities, our proposed polarimetric feature accurately captures the geometric structure of the target, such as the surface scatterer and diplane scatterer. This establishes a clear correspondence between the feature and the physical structure, laying a solid foundation for SAR target recognition.

2) Target Recognition Performance: To verify the effectiveness of the proposed polarimetric feature, the A-ConvNets is employed as the network backbone. This feature is used as the input, and its recognition performance is compared with those of CV-3D-CNN [29] and MS-CVNN [47]. Y4O decomposition [34], [35], as a typical polarization decomposition

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TABLE IV

ABLATION STUDY ON THE SCATTERER SELECTION USING A-CONVNETS AS NETWORK BACKBONE. METRICS (%) INCLUDING OA, AA, AND KAPPA ARE USED FOR EVALUATION. THE BEST RESULTS ARE EMPHASIZED IN BOLD

Input	Scatterer selection and feature normalization						Metrics		
	surface	*diplane*	*diplane*-45°	*dipole*	*helix*	*normalization*	OA	AA	Kappa
$r_{surface}, r_{diplane}, r_{diplane45°}$	✓	✓	✓				91.33	91.33	89.81
$r_{surface}, r_{diplane}, r_{dipole}$	✓	✓		✓			95.54	95.54	94.81
$r_{surface}, r_{diplane}, r_{helix}$	✓	✓			✓		93.68	93.68	92.25
$r_{surface}, r_{diplane}, r_{dipole}, r_{helix}$	✓	✓		✓	✓		**98.78**	**98.78**	**98.51**
$p_{surface}, p_{diplane}, p_{dipole}, p_{helix}$	✓	✓		✓	✓	✓	56.35	56.35	50.97

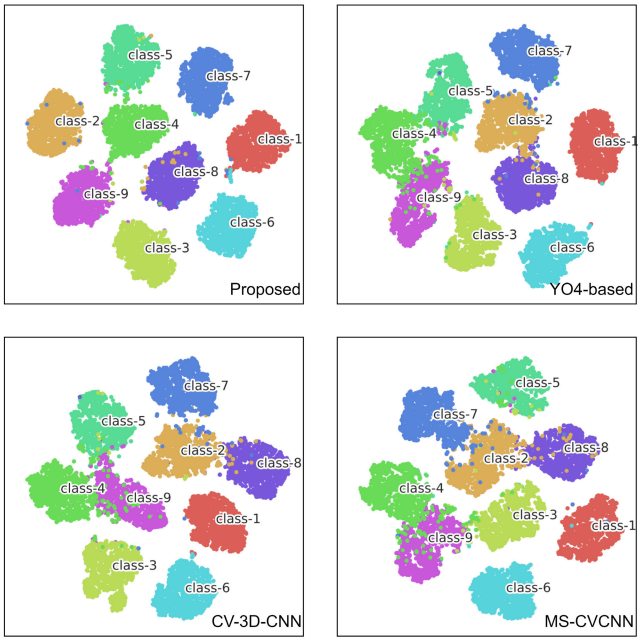


Fig. 15. Output feature visualization of the test methods on the test set of GOTCHA, using t-sne [50].

method, is also used for feature extraction and then compared with the proposed feature. The Pol-MSTAR synthetic datasets, with different SCRs of 20 and 10 dB, alongside the real dataset GOTCHA, served as benchmark datasets. The comparison of recognition accuracy is depicted in Table III. It becomes evident that the proposed feature delivers comparable performance with CV-3D-CNN and MS-CVNN under high SCR conditions. Yet, it surpasses them under low SCR conditions. Furthermore, it outperforms the reference methods when applied to the real image dataset, GOTCHA. These recognition results validate the effectiveness of the proposed feature.

A comprehensive confusion matrix for the proposed feature in the GOTCHA dataset is shown in Fig. 14. Both Classes 1 and 6 have almost achieved 100% recall rates, while the misclassification probability for other classes toward Classes 1 and 6 remains marginal. These results are closely associated with the physical geometry of the targets. Class 1 and 6 targets exhibit significantly different structures compared to the other classes, which predominantly consist of small cars with subtle structural variations. Despite these subtle differences, the proposed polarimetric feature aptly describes the structural differences among the classes. It also enables refined classification through the A-ConvNets, thereby indirectly demonstrating the effectiveness of the proposed feature.

3) Ablation Study: To verify the plausibility of our features in terms of scatterer selection and feature normalization, we perform ablation experiments. The A-ConvNets is used as a network backbone for five different types of inputs. Ablation experimental results are presented in Table IV. Comparing the first three rows of Table IV, we can see that the correlation feature with the dipole and helix maintain considerable advantages over the correlation feature with the diplane (45° orientation angle), which is consistent with our assertion in Section III-D. Hence, we combine the four typical scatterers together, extract correlation features with them, and achieve the best recognition performance. Comparing the last two rows of Table IV, we can see that feature normalization drastically degrades target recognition performance. Normalizing features may result in the loss of intensity information pertaining to the target, causing it to blend into the background (which can be easily observed in Fig. 12), thereby severely interfering with target recognition.

4) Feature Visualization and Analysis: Besides the outperformance on evaluation metrics of the proposed method, we want to take a further discussion and explanation of the results in this section. We use t-sne [50] to visualize the output features of test methods on the test set of GOTCHA, shown in Fig. 15. Dots of different colors represent samples of different categories in Fig. 15. We can see that all categories of samples are more clearly distinguished in the proposed method, except for a few mistaken samples mixed in the other clusters. And the feature boundary between almost all categories is much clearer in the proposed method. For instance, the boundaries between classes 4, 5, and 9 in Fig. 15(b) and (c) are less clear; the boundaries between classes 2 and 8 are not clear in Fig. 15(b)–(d). Similar results can also be observed on Pol-MSTAR -10 and -20 dB and will not be repeated here.

To demonstrate the discrimination performance of the proposed feature, we use center distances [51] for quantitative assessment. The intraclass center distance is defined as

$$L_{\text{intra},i} = \frac{1}{N_i} \sum_{n=1}^{N_i} \left\| \boldsymbol{x}_{i,n} - \boldsymbol{c}_i \right\|_2^2 \tag{24}$$

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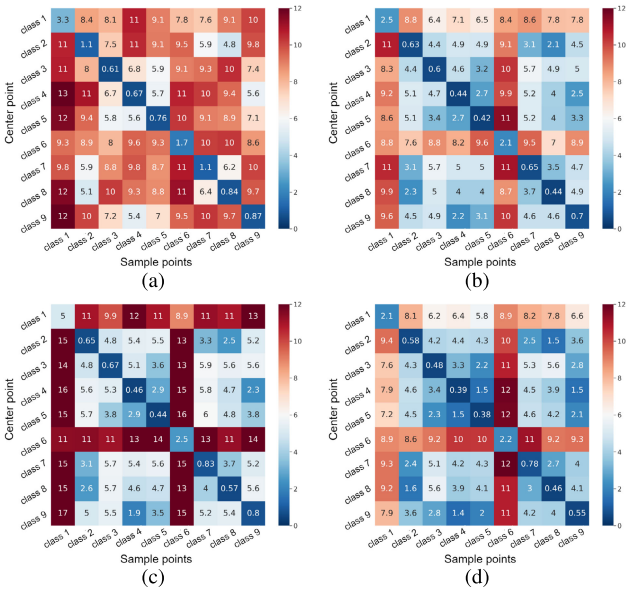


Fig. 16. Distance matrices of the extracted features for the (a) proposed method, (b) YO4-based method, (c) CV-3D-CNN, and (d) MS-CVCNN. It is preferable for the diagonal elements to be minimized (for intraclass compactness) while maximizing the off-diagonal elements (for interclass separation).

where $L_{\text{intra},i}$ denotes the center distance for class i, $\boldsymbol{x}_i, n$ denotes the nth sample of the ith class, $\boldsymbol{c}_i$ denotes the sample center for class i, and N_i is the number of the ith class samples. The interclass center distance from class i (sample center) to class j (samples) is defined as

$$L_{\text{inter},i\to j} = \frac{1}{N_j}\sum_{n=1}^{N_j}\left\|\boldsymbol{x}_{j,n} - \boldsymbol{c}_i\right\|_2^2. \tag{25}$$

We develop a distance matric $\mathcal{L}$ by combining the intra- and interclass distances

$$\mathcal{L}_{ij} = \begin{cases} L_{\text{intra},i} & \text{if} \quad i = j \\ L_{\text{inter},i\to j} & \text{otherwise.} \end{cases} \tag{26}$$

Fig. 16 shows the distance matrices of the extracted features for (a) the proposed method, (b) the YO4-based method, (c) the CV-3D-CNN and (d) the MS-CVCNN. It is preferable for the diagonal elements to be minimized (for intraclass compactness) while maximizing the off-diagonal elements (for interclass separation). It can be easily seen that the samples of classes 1 and 6 are far away from the samples of the other classes. This is consistent with the optical sample appearances shown in Fig. 7. However, for similar classes, such as classes 3 and 7 as well as classes 4 and 5, the interclass distances of the feature for the proposed method are significantly larger than those for the YO4-based method, the CV-3D-CNN, and the MS-CVCNN. For instance, the inter-class distance from classes 4 to 5 in the proposed method is about 5.6, while those in the YO4-based method, the CV-3D-CNN, and the MS-CVCNN are about 2.7, 2.9, and 1.5, respectively. The distance matrix for the proposed method shows the characteristics of extremely small diagonal elements and large off-diagonal elements, demonstrating the advantages of intraclass compactness and interclass separation.

TABLE V
CONTRASTIVE CENTER LOSS COMPARISON

	Proposed	YO4-based	CV-3D-CNN	MS-CVCNN
Loss	**0.07388**	0.07711	0.07503	0.07595

Besides the distance matrix, we also use another indicator, contrastive-center loss [52] to evaluate the discrimination performances, which is defined as

$$\text{Loss} = \frac{1}{2}\sum_{i=1}^{C}\frac{L_{\text{intra},i}}{\sum_{j\neq i} L_{\text{inter},j\to i}}. \tag{27}$$

Table V lists the contrastive center losses for the test method, among which the proposed method performs the best. From all the above, we can conclude that the proposed method in this article can better capture the differences between categories, achieve intracategory compactness, and achieve intercategory separation, outperforming the reference methods significantly.

V. CONCLUSION

This article investigates the contribution of polarimetric information to SAR target recognition, introducing a novel visualization approach that underscores the significance of polarimetric channels by using a channel-wise convolutional structure as the polarimetric channel proxy and adapting Grad-CAM. For the first time, we offer a quantitative evaluation and analysis of the impact of polarimetric information, achieved through comparative experiments and our proposed approach. We demonstrate again that: 1) polarimetric information can effectively mitigate the interference of clutter, significantly surpassing single-polarization intensity information and 2) within the polarimetric covariance matrix elements, the cross-polarization intensity and the correlations between cross-polarization and co-polarization are pivotal components.

Inspired by these conclusions, we introduce a polarimetric correlation feature. This feature can accurately delineate the geometric structure of targets, such as surface scatterers and diplane scatterers, thereby establishing a distinct correspondence between the feature and the physical structure. Experimental results on simulated and real datasets demonstrate that this method significantly outperforms the existing comparison methods. This provides a strong foundation for SAR target recognition. Future research will further explore the potential of polarimetric information in SAR target recognition from the perspective of polarimetric scattering characteristics.

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